

AI-Powered Mobile Platform for Democratizing Sports Talent Assessment

1. Problem Statement

Traditional sports talent identification systems are often limited to urban regions, professional academies, and expensive infrastructure. Many talented athletes from rural and underserved areas do not receive proper opportunities due to lack of access, trained evaluators, and standardized assessment systems. Current talent assessment methods are mostly manual, time-consuming, biased, and difficult to scale nationwide. There is a strong need for a smart, affordable, and accessible platform that can evaluate sports talent fairly using Artificial Intelligence and mobile technology.

2. Proposed Solution

The proposed system is an AI-powered multimodal mobile platform designed to assess sports talent using text, image, video, and audio inputs. The platform enables athletes to upload their performance data through a smartphone, allowing AI models to evaluate fitness, agility, stamina, posture, and sport-specific skills. The application uses Computer Vision, Machine Learning, Speech Processing, and Pose Detection techniques to provide fair and scalable talent assessment. The platform supports multiple sports such as cricket, football, kabaddi, athletics, and basketball. The system aims to democratize sports talent discovery by providing equal opportunities to students and athletes from all regions, including rural communities.

3. Multimodal Input Features

- Video Analysis – Detects movement, posture, agility, sprint speed, and sport-specific actions.
- Image Processing – Analyzes body posture, stance, and physical alignment.
- Audio Processing – Supports multilingual voice commands and spoken feedback.
- Text Input – Collects athlete profiles, fitness records, and performance history.

4. System Workflow

- 1 User Registration – Athlete creates a profile with personal and sports information.
- 2 Data Upload – User uploads video, image, audio, or text-based inputs.
- 3 AI Processing – Machine Learning models analyze movements, posture, and performance.
- 4 Performance Evaluation – System calculates scores for strength, endurance, speed, agility, and skills.
- 5 Sport Recommendation – AI recommends suitable sports based on athlete abilities.
- 6 Feedback Generation – Personalized training suggestions and improvement tips are provided.
- 7 Talent Dashboard – Coaches and organizations can identify top-performing athletes.

5. Technologies Used

Component	Technology
Mobile App	Flutter / React Native / Android
Backend	Python / Node.js

AI & Machine Learning	TensorFlow / PyTorch
Pose Detection	MediaPipe / OpenPose
Database	Firebase / MongoDB
Cloud Services	AWS / Google Cloud

6. Advantages of the System

- Affordable and accessible talent assessment.
- Supports rural and underserved communities.
- Reduces bias through AI-based evaluation.
- Scalable for schools, academies, and government programs.
- Provides personalized training recommendations.
- Enables nationwide sports talent discovery.

7. Conclusion

The AI-Powered Mobile Platform for Democratizing Sports Talent Assessment provides a modern and scalable approach to identifying and nurturing sports talent. By integrating multimodal AI technologies such as video analysis, image processing, audio interaction, and text-based profiling, the platform can create equal opportunities for athletes across different regions and economic backgrounds. This solution can significantly improve grassroots sports development and help discover hidden talent through intelligent and data-driven assessment.